ABHISHEK DIXIT

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Profile

Embedded Software & Cybersecurity Engineer with 10+ years of experience in automotive ECU software development, RTOS platforms, low-level design, and cybersecurity implementation. Proven expertise in Embedded C, Secure Boot, HSM integration, secure diagnostics, UDS/CAN protocols, and AUTOSAR-based systems. Experienced in implementing ECU security features and secure communication mechanisms aligned with ISO/SAE 21434, while working within automotive SDLC processes and regulatory environments including UNECE WP.29 and CSMS requirements.

Skills

- Secure Diagnostics, Secure Boot, Secure Flash, SecOC, Secure Debug, Secure Storage, Secure Event Logging, Secure Key Management.
- Cryptography methods, PKI, Threat Analysis and Risk Assessment (TARA) using STRIDE, SDLC.
- AUTOSAR Crypto/Diagnostic/COM Stack, MCAL, HSM.
- ISO/SAE 21434, UNECE WP.29 (155 CSMS and 156 SUMS), NIST, ISO27001 Annex A controls, ASPICE and ISO 26262
- OP-TEE Trusted Applications for secure execution in Linux-based systems and RTOS.
- UDS protocol (ISO 14229), CAN, Bootloader, TCP/IP stack.
- Designing of system and software requirements using DOORS and Polarion
- JIRA, Confluence, BCT, MKS, GIT, Tortoise SVN, Lauterbach Trace32, MULTI, WinIdea, UDE, EB Tresos, Enterprise Architect, Medini.
- Embedded C, C++, MISRA C Standards, Static Code Analysis. .
- Ambarella CV22 (SoC - Linux), Renesas (F1x, P1x), Infineon TC364 and TC277, and ST Microelectronics SPC584B70.

Experience

Security Consultant at T&S Services GmbH, Stuttgart Germany

Aug 2023 – Jun 2025

Ford Smart Vision and SonceBoz EHPS Projects.

- Coordinated security requirements and defined security concepts together with customer teams.
- Developed complete security concepts including bootloader security, anti-tampering, and EOL security.
- Implemented Secure Boot, Secure Access, Secure Flash, Key Injection, and debugging strategies for multi-platform ECUs (TriCore, Ambarella CV22, ST).
- Designed anti-tampering concepts, secure diagnostics, secure debug, and secure storage mechanisms.
- Integrated security and diagnostic functions across Linux and RTOS cores using IPC and OP-TEE Trusted Applications.
- Developed OP-TEE Trusted Applications to secure cryptographic operations and critical system functions.
- Implemented secure DCF/UCB flashing mechanisms leveraging ISO 14229 diagnostics and MCU-specific programming sequences for protected configuration updates.
- Created and maintained comprehensive ISO/SAE 21434 cybersecurity documentation, including technical security specifications, static and dynamic security architectures and compliance evidence across multi-platform ECUs.
- Created Low-Level Designs (LLD) and software requirements for security modules (Secure Boot, Secure Flash) and performed unit testing, peer code reviews, and static analysis following MISRA C guidelines.

Cybersecurity Architect at Continental, Bengaluru India — Sep 2022 – Jun 2023

Worked as Security Manager and Security Architect for the Daimler DCDC48V Battery Management Project.

- Managed the security lifecycle (SYS.1–SYS.3) and aligned CSIA and OEM security requirements.
- Led the creation and maintenance of ISO/SAE 21434 work products, including Item Definition, Security Plan, Vulnerability Analysis (VARA), Security Case, Security Concept, and Reuse Analysis.
- Performed comprehensive security risk analyses, including TARA (STRIDE), third-party evaluations, and external component assessments, to identify threats, define security goals, and derive technical security requirements across the full ECU lifecycle.
- Coordinated cross-functional teams and supported assessments and audits.
- Ensured compliance with ASPICE and ISO 26262, conducted security assessments for release readiness.
- Configured and managed the Key Management System (KMS) for secure production and commissioning processes.

Automotive Software Developer at Bosch, Bengaluru India — Feb 2016 – Aug 2022

Worked on multiple Bosch projects (RADAR, VMPS, USS) for various OEM customers. and was deputed to the Bosch Abstatt site for the VMPS project.

- Coordinated diagnostic and security requirements with OEM teams and participated in architecture workshops for embedded control units.
- Developed and integrated embedded software in C environment for AUTOSAR Crypto Stack, DCM, and CAN communication with UDS (ISO 14229) protocols, adhering to SDLC and quality processes.
- Implemented security-relevant functions such as Secure Access, RNG generation, secure flash, secure timeout handling, ESCRYPT/BCL cryptographic functions integration, and diagnostic services according to UDS (ISO 14229).
- Developed and integrated DCM/NVM-based diagnostic functions, including radar alignment, lens heating, and UDS-on-CAN support.
- Implemented J1939 diagnostic tunneling and created DBC files for GNSS, IMU, and sensor-fusion signals.
- Configured the AUTOSAR COM Stack for CAN communication in high-performance ECUs.

Software Developer at KPIT, Bengaluru India — Oct 2012 – Jan 2016

Developed MCAL Low-Level Drivers Development for Renesas microcontrollers.

- Developed low-level drivers (MCAL ADC) in Embedded C/RTOS environment according to SDLC processes, conducting unit testing, code inspections, and hardware validation.
- Analyzed, implemented, and tested customer-specific and AUTOSAR-based ADC driver requirements.
- Validated driver performance in collaboration with hardware teams.

Education

TÜV SÜD South Asia Pvt. Ltd. — India
Certified Automotive Cyber Security Practitioner (CACSP) — Jun 2024

Jaya Engineering College — India
Bachelor of Engineering, Electrical & Electronics Engineering — Jun 2006 – Jun 2010

Language

- English C1
- German A2
- Hindi Native